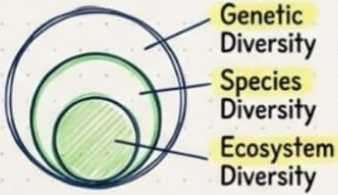
Chapter 13: Biodiversity & Conservation

The Master Study Guide

I. Introduction to Biodiversity

Defined as the variety of life on Earth at all levels: genetic, species, & ecosystem. Critical for ecosystem stability & services. Global vs. Local patterns.



II. Threats: The Evil Quartet

- Habitat Loss & Fragmentation (Greatest threat!)
- Overexploitation (Hunting, harvesting)
- Invasive Species (Competes/predates native spp.)
- Climate Change (Disrupts habitats & life cycles)

Fig. 1: *Rauwolfia vomitoria* leaf (Anti-hypertensive, antipsychotic)

III. Conservation Strategies

Focuses on protecting & restoring biodiversity. In-situ vs. Ex-situ.

Strategy	Examples
In-situ (On-site)	Protected Areas, Biosphere Reserves, National Parks, Sanctuaries
Ex-situ (Off-site)	Botanical Gardens, Zoos, Seed Banks, Gene Banks, Cryopreservation

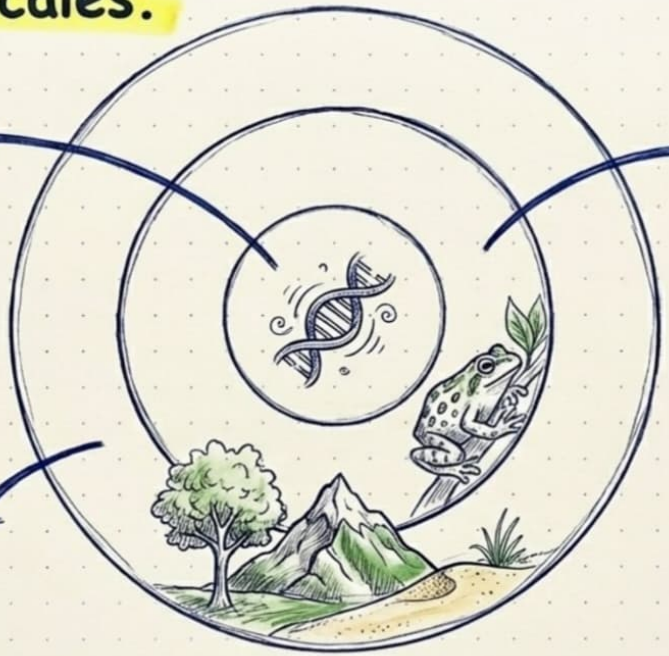
IV. International Efforts

IUCN Red List (Threatened Species)
CBD (Convention on Biological Diversity)
CITES (Trade regulation)

Exam focus: Macro-trends, The Evil Quartet, Conservation Matrices.

Biodiversity exists at three nesting scales.

Genetic Diversity.
Single species showing high diversity.
Ex: *Rauwolfia vomitoria* in Himalayas (varies in active chemical reserpine). India has 50k rice strains, 1k mango varieties.



Species Diversity.
Variety between species in a region.
Ex: Western Ghats have greater amphibian diversity than Eastern Ghats.

Ecological Diversity.
Ecosystem level.
Ex: India (deserts, rain rain forests, coral reefs) has greater ecosystem diversity than Norway.

If an alien visited Earth, the numbers would baffle them.

>20,000

species of ants

nearly

20,000

species of orchids

28,000

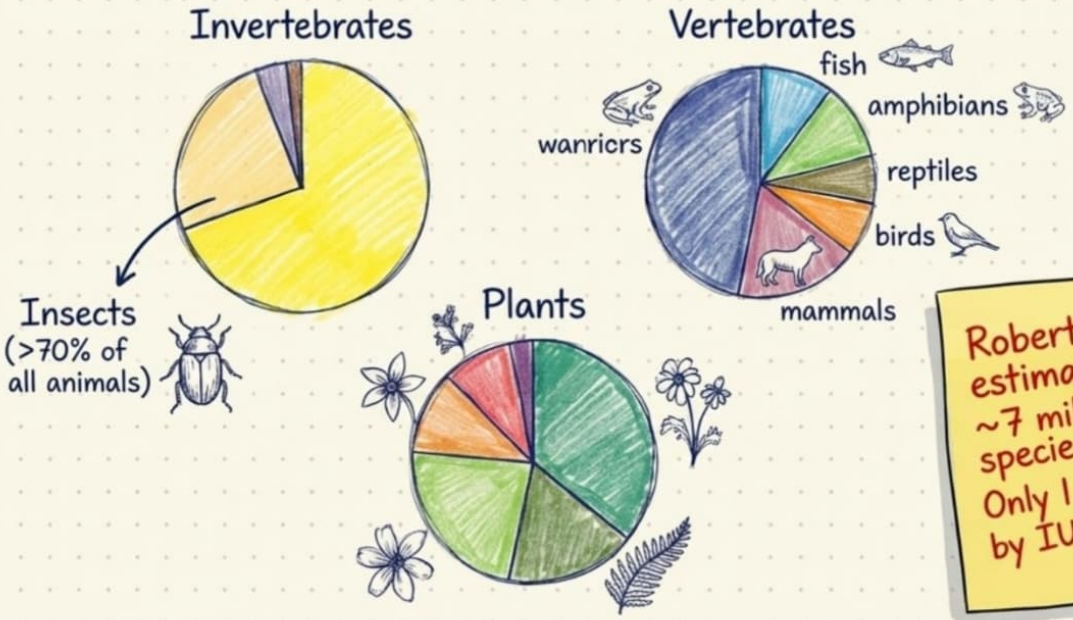
species of fishes

3,00,000

species of beetles

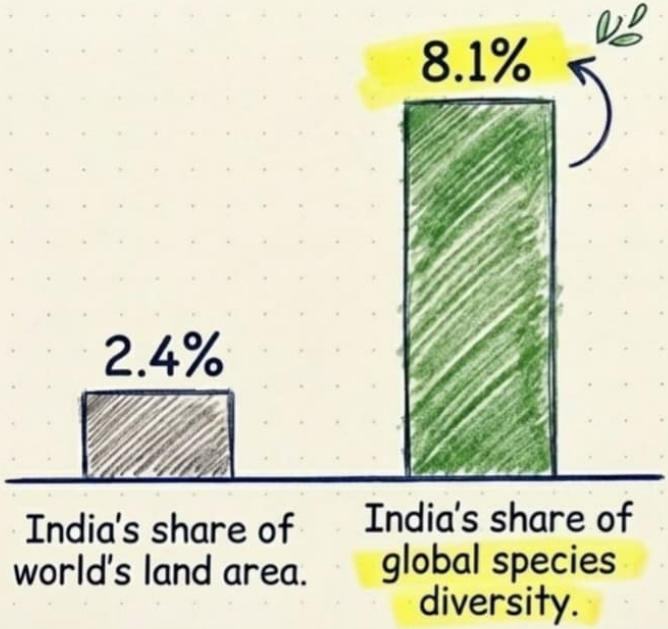
Biodiversity: The term popularized by sociobiologist Edward Wilson to describe combined diversity at all levels of biological organization.

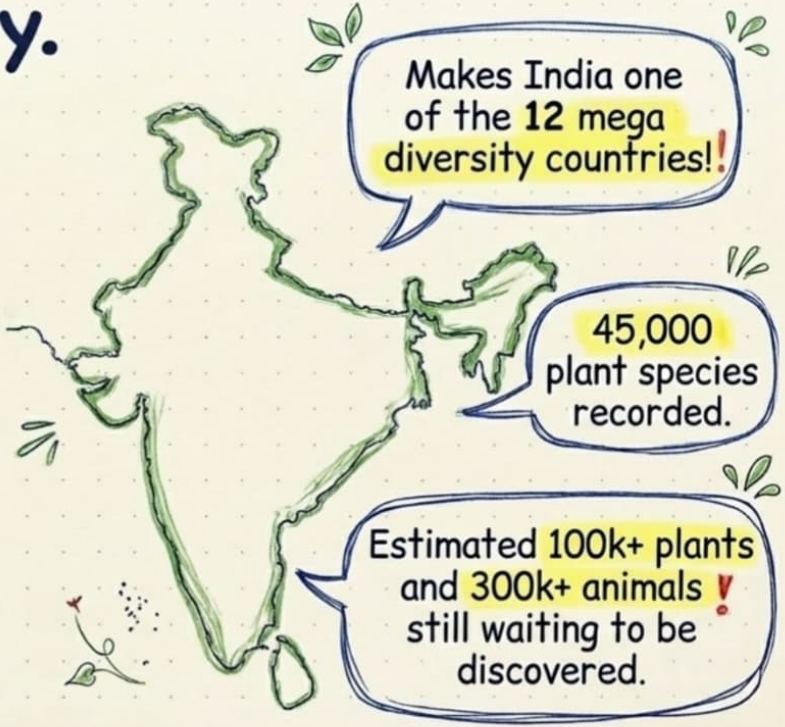
Animals dominate the global inventory, and insects dominate the animals.



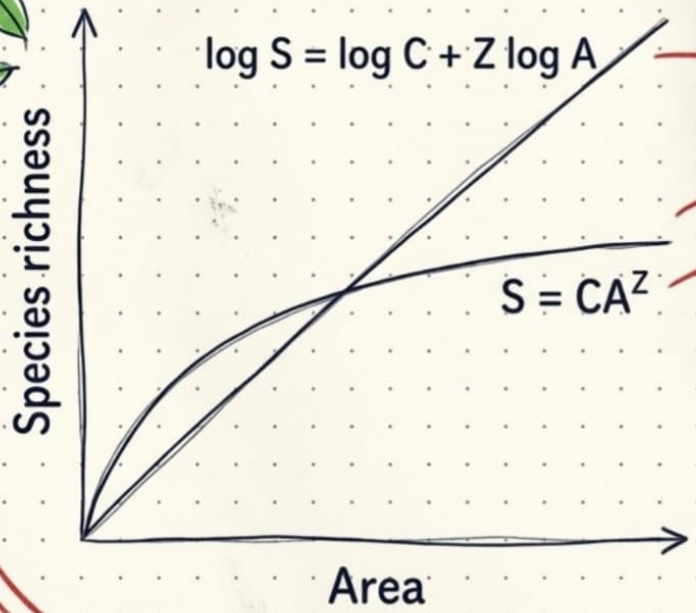
Robert May's estimate:
~7 million global species.
Only 1.5M described by IUCN (2004)!

India punches way above its weight in global biodiversity.





Alexander von Humboldt's Math of Biodiversity



Student Annotations

S = Species richness

A = Area

C = Y-intercept

Z = Slope of the line.

- Usually 0.1 to 0.2 regardless of region/taxa!
- ⚠ But jumps to 0.6 to 1.2 for entire continents.

Life gets richer as you move toward the equator.

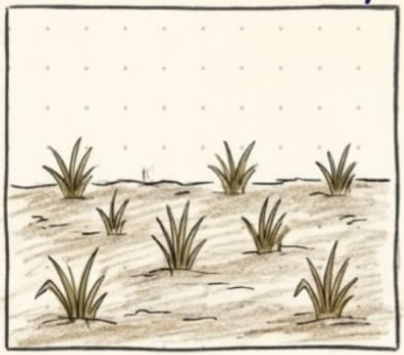
Why are the tropics so special?

1. **More Time:** No frequent glaciations; millions of years of undisturbed evolution.
2. **Constant Environment:** Less seasonal, promoting niche specialization.
3. **More Solar Energy:** Higher productivity leads to higher diversity.

The 'Rivet Popper Hypothesis' (Paul Ehrlich).

David Tilman's plots proved that diversity creates stability.

Plot A (Low Diversity)



- High year-to-year variation in biomass.
- Vulnerable to alien invasions.

Plot B (High Diversity)

- Less year-to-year variation.
- Increased productivity.
- Resilient to occasional disturbances.

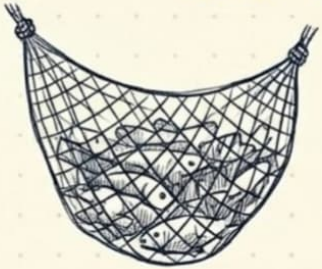
Conclusion: Rich biodiversity is imperative for the survival of the human race, not just ecosystem health.

The 'Evil Quartet' driving species loss.

Habitat Loss & Fragmentation

The primary driver. Amazon rain forest ('lungs of the planet') cut for soya beans & cattle. Large territories broken into small fragments.

Over-exploitation



'Need' turns to 'greed.' Extinctions of Steller's sea cow & passenger pigeon.

Alien Species Invasions

Nile perch wiped out 200+ cichlid fish in Lake Victoria.

Weeds like Water Hyacinth and Lantana.

Co-extinctions

Obligatory associations. Extinction of a host fish kills its specific parasites.

Plant-pollinator mutualism breakdown.

We are currently driving the 'Sixth Extinction'.

5 episodes of mass extinction occurred before humans.
But the current rate is 100 to 1,000 times faster!

IUCN 2004 Losses
(Last 500 Yrs)

784 total species lost.

Dodo
(Mauritius)

Quagga
(Africa)

Dodo
(Mauritius)

Thylacine
(Australia)

Steller's Sea Cow
(Russia)

Currently facing threat:

- 12% birds
- 23% mammals
- 32% amphibians
- 31% gymnosperms

Three arguments for why we must conserve.

Narrowly Utilitarian

- Direct economic benefits.
- Food, firewood, industrial products.
- 25% of all drugs derive from plants.
- Bioprospecting.

Broadly Utilitarian

- Ecosystem services.
- Amazon produces 20% of Earth's oxygen.
- Natural pollination by bees/birds.
- Aesthetic pleasures (can't put a price tag on a bulbul's song).

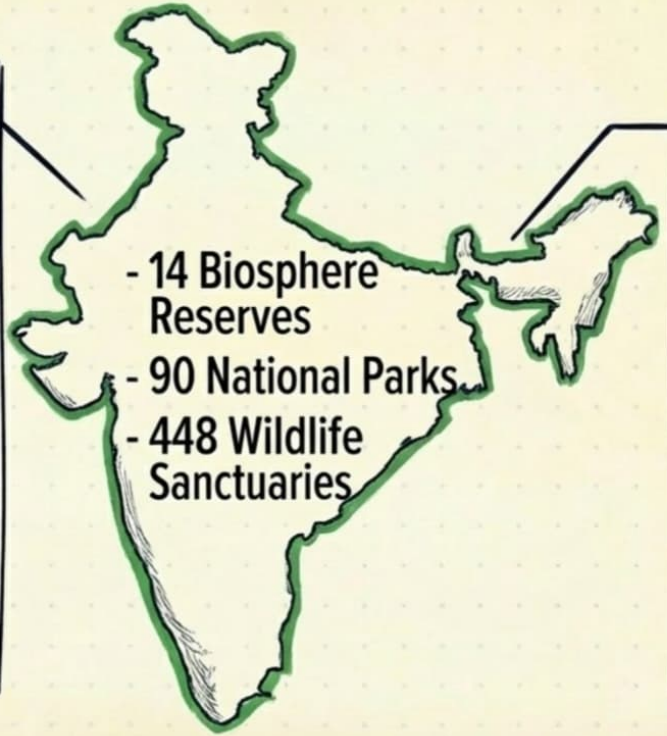
Ethical

- Intrinsic value of every species.
- We owe it to millions of microbes, plants, and animals.
- Moral duty to pass our biological legacy to future generations.

Protecting India's biological wealth on the ground.

Biodiversity Hotspots: Regions with high richness and endemism (species found nowhere else).

34 globally → 3 cover India (Western Ghats & Sri Lanka, Indo-Burma, Himalaya).



Sacred Groves: Cultural tradition of atrection of total protection. Found in Khasi/Jaintia Hills (Meghalaya), Aravalli (Rajasthan), Western Ghats.

Last refuge for rare plants!

How we conserve: In situ vs. Ex situ.

	In situ (On site)	Ex situ (Off site)
Approach	Protect the whole ecosystem in its natural habitat.	Take threatened species out of natural habitat for special care.
Philosophy	'Save the entire forest to save the tiger.'	Urgent measures for species facing high risk of extinction in the wild.
Methods ✓	Biosphere reserves, National parks, Sanctuaries, Biodiversity hotspots. ✓	Zoos, botanical gardens, cryopreservation of gametes, seed banks, *in vitro* fertilization. ✓

Biodiversity knows no political boundaries.

1992: The Earth Summit (Rio de Janeiro).
All nations called to conserve biodiversity and sustainably use its benefits.

2002: World Summit on Sustainable Development (Johannesburg).
190 countries pledge to significantly reduce current rate of biodiversity loss by 2010.

Conservation is a **collective responsibility.** Pass it on.